Prompt optimisation

OPRO

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GSM8K: grade-school maths word problems

Q: Alannah, Beatrix, and Queen are preparing for the new school year and have been given books by their parents. Alannah has 20 more books than Beatrix. Queen has 1/5 times more books than Alannah. If Beatrix has 30 books, how many books do the three have together?

A: The instruction goes here Then the working, then a number

At the start of the answer

The answer on file

140

Right if they match



7,473 training problems

To optimise on (a 3.5% sample)

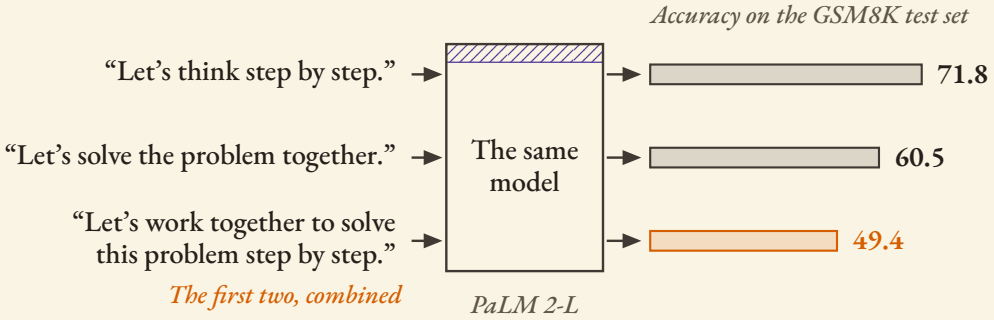


1,319 test problems

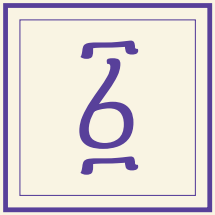
To check the result on

Yang et al., Figure 3 and section 5

Same model, different instruction



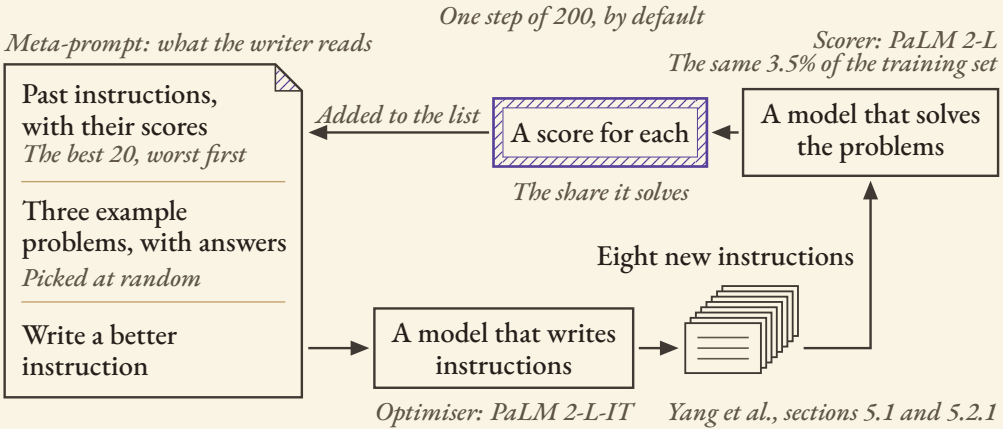
Yang et al., Large Language Models as Optimizers, section 5.2.3



Section 1

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One step of OPRO



What the meta-prompt looks like

Yang et al., Figure 3

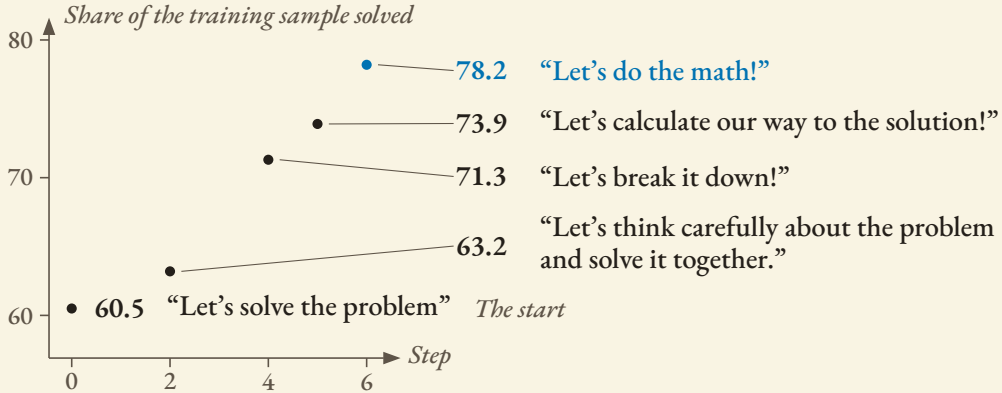
[Some texts and their scores, lowest score first]
text: Let's figure it out!
score: 61
text: Let's solve the problem.
score: 63
(. . . more instructions and scores . . .)
[How a text is used: it replaces <INS>]
input: Q: Alannah, Beatrix, and Queen [. . .] how many books do the three have together?
A: <INS>
output: 140
(. . . more exemplars . . .)
[Write a new text, unlike the old ones, with a score as high as possible]

The instructions so far, worst first
Worst first beat best first and random order

Example problems, with the slot marked

The request

Better instructions within six steps



Yang et al., section 5.2.1, PaLM 2-L solving, PaLM 2-L-IT writing

Fifty at once against eight a step



Fifty new instructions, written in one go

It never sees their scores

Eight a step, each step reading the scores so far

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Accuracy on the test set

60.8 Fifty at once: “Let’s solve the problem”, the start itself

76.3 OPRO, within six steps: “Let’s do the math!”

Yang et al., section 5.3

Eight instructions on GSM8K, ranked

Share of the GSM8K test set solved by PaLM 2-L, the instruction at the start of its answer

🏆	80.2	**"Take a deep breath and work on this problem step-by-step."**	*Found by PaLM 2-L-IT*
⑨	79.9	"Break this down."	*Found by PaLM 2-L*
⑧	78.5	"A little bit of arithmetic and a logical approach will help us quickly arrive at the solution to this problem."	*Found by gpt-3.5-turbo*
⑦	74.5	"Let's combine our numerical command and clear thinking to quickly and accurately decipher the answer."	*Found by gpt-4*
⑥	71.8	"Let's think step by step."	*Baseline, Kojima et al.*
⑤	60.8	"Let's solve the problem."	*Baseline*
④	58.8	"Let's work this out in a step by step way to be sure we have the right answer."	*Baseline, Zhou et al.*
③	34.0	" " *No instruction at all*	*Baseline*